

Single Neuron $y = w_1x_1 + w_2x_2 + b$ $\text{activation} = \sigma(y)$ $\text{empirical risk} = \frac{1}{n} \sum_{i=1}^n \ell(y_i, \hat{y}_i)$ ℓ_2 Loss for regression: $\frac{1}{2} \ y - \hat{y}\ ^2$ ℓ_1 Loss for classification: $\sum_{i=1}^n \max(0, 1 - y_i \hat{y}_i)$ Neuron types Input Neuron: x_i Hidden Neuron: h_j Output Neuron: y_k Connections Weighted Connections: w_{ij} Bias: b_j Activation Functions Sigmoid: $\sigma(x) = \frac{1}{1 + e^{-x}}$ ReLU: $\max(0, x)$ Tanh: $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ Leaky ReLU: $\max(0, x)$ if $x > 0$, else αx Softmax: $\sigma(x) = \frac{e^{x_i}}{\sum_j e^{x_j}}$ Loss Functions Cross Entropy: $\mathcal{L} = -\sum_i y_i \log(\hat{y}_i)$ Mean Squared Error: $\mathcal{L} = \frac{1}{2} \sum_i (y_i - \hat{y}_i)^2$ Backpropagation Forward Pass: Calculate $\hat{y}$ from x Backward Pass: Calculate gradients $\frac{\partial \mathcal{L}}{\partial w}$ Weight Update: $w_{ij} = w_{ij} - \eta \frac{\partial \mathcal{L}}{\partial w_{ij}}$ Neural Network Architectures Feedforward Neural Network (FNN): Simple linear layers. Convolutional Neural Network (CNN): Local receptive fields, shared weights. Recurrent Neural Network (RNN): Temporal dependencies.	11-12) Backpropagation - Backward pass: calculate gradients $\frac{\partial \mathcal{L}}{\partial w}$ - Weight update: $w_{ij} = w_{ij} - \eta \frac{\partial \mathcal{L}}{\partial w_{ij}}$ 13-14) Forward Pass - Input x is passed through layers - Output $\hat{y}$ is calculated 15-16) Loss Calculation - Loss $\mathcal{L}$ is calculated - Gradients $\frac{\partial \mathcal{L}}{\partial w}$ are backpropagated 17-18) Weight Update - Weights w are updated 19-20) Model Evaluation - Model is evaluated on test data 21-22) Hyperparameter Tuning - Hyperparameters are tuned 23-24) Model Deployment - Model is deployed to production 25-26) Model Monitoring - Model performance is monitored 27-28) Model Retraining - Model is retrained on new data 29-30) Model Archiving - Model is archived for future use 31-32) Model Versioning - Model versions are tracked 33-34) Model Documentation - Model documentation is maintained 35-36) Model Security - Model security is ensured 37-38) Model Compliance - Model compliance is verified 39-40) Model Transparency - Model transparency is promoted 41-42) Model Explainability - Model explainability is enhanced 43-44) Model Interpretability - Model interpretability is improved 45-46) Model Accountability - Model accountability is established 47-48) Model Responsibility - Model responsibility is assigned 49-50) Model Ethics - Model ethics are considered 51-52) Model Fairness - Model fairness is ensured 53-54) Model Bias - Model bias is identified and mitigated 55-56) Model Discrimination - Model discrimination is prevented 57-58) Model Harassment - Model harassment is stopped 59-60) Model Stalking - Model stalking is monitored 61-62) Model Cyberstalking - Model cyberstalking is prevented 63-64) Model Identity Theft - Model identity theft is prevented 65-66) Model Financial Fraud - Model financial fraud is prevented 67-68) Model Intellectual Property Theft - Model intellectual property theft is prevented 69-70) Model Trade Secret Theft - Model trade secret theft is prevented 71-72) Model Patent Infringement - Model patent infringement is prevented 73-74) Model Copyright Infringement - Model copyright infringement is prevented 75-76) Model Trademark Infringement - Model trademark infringement is prevented 77-78) Model Unfair Competition - Model unfair competition is prevented 79-80) Model False Advertising - Model false advertising is prevented 81-82) Model Deceptive Marketing - Model deceptive marketing is prevented 83-84) Model Unfair Business Practices - Model unfair business practices are prevented 85-86) Model Anticompetitive Behavior - Model anticompetitive behavior is prevented 87-88) Model Monopolistic Behavior - Model monopolistic behavior is prevented 89-90) Model Restrictive Trade Practices - Model restrictive trade practices are prevented 91-92) Model Unfair Contract Terms - Model unfair contract terms are prevented 93-94) Model Unfair Dispute Resolution - Model unfair dispute resolution is prevented 95-96) Model Unfair Arbitration - Model unfair arbitration is prevented 97-98) Model Unfair Mediation - Model unfair mediation is prevented 99-100) Model Unfair Conciliation - Model unfair conciliation is prevented	35) Introduction RNN - Input x_t is passed through hidden state h_t - Output y_t is calculated 36) RNN Architecture - Hidden state h_t is updated 37) RNN Variants Vanilla RNN: Simple recurrent connections. Long Short-Term Memory (LSTM): Memory cells, forget gates. Gated Recurrent Unit (GRU): Reset and update gates. 38) RNN Applications Text Generation: Predict next word. Machine Translation: Translate between languages. Speech Recognition: Convert speech to text. Image Captioning: Generate captions for images. Video Classification: Classify video content. Music Generation: Generate musical sequences.
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